

Smart contract architecture

Contract map for the markets stack: ERC-4626 PairVaults, MarketLendingModule, adapters, oracle, rates, CRE liquidation, and how funds move — with repo references.

1. Overview

Lenders supply **Circle USDC** into isolated per-pair **ERC-4626 PairVaults**. Borrowers lock a **DEX V3 or V4 LP NFT** and borrow USDC from the matching pair. Interest accrues on-chain. Underwater loans are liquidated by the market owner or **Chainlink CRE** via `CreliquidationReceiver`. After debt is pushed back to the vault, a protocol fee may go to `feeTo`.

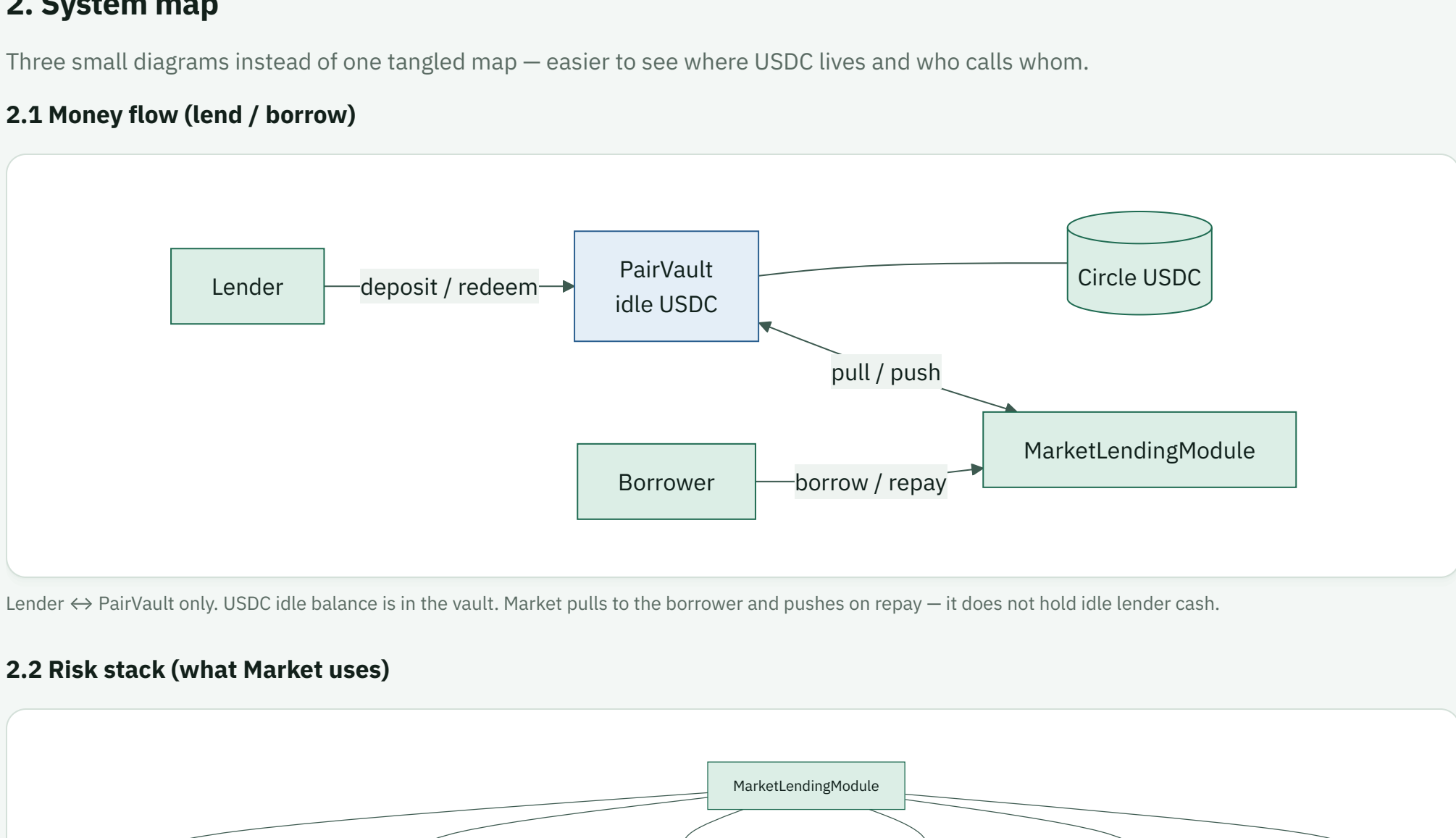
Idle USDC lives in **PairVault**, not on the market. `Legacy lendUsdc / withdrawLender revert ( UsePairVault )`.

Canonical address file: `smart-contracts/deployments/sepolia-markets.json` . Repo: github.com/unblocktechie/LPFy.

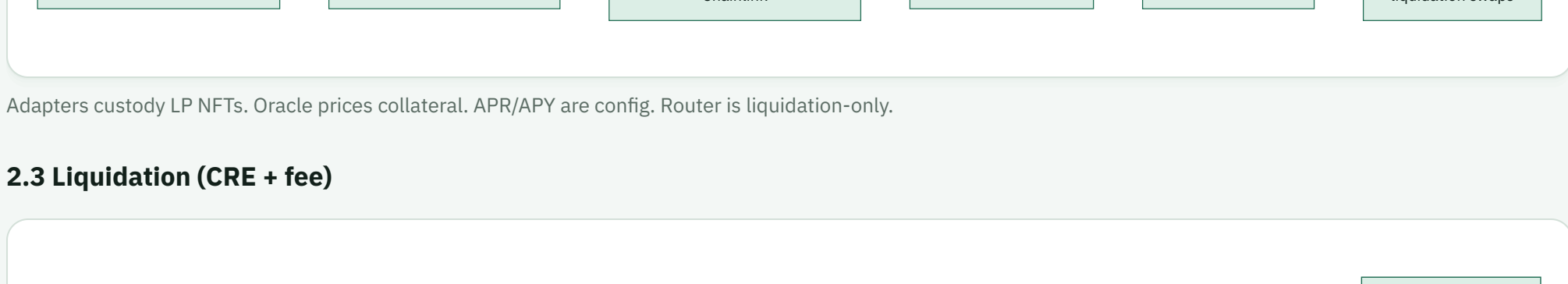
2. System map

Three small diagrams instead of one tangled map — easier to see where USDC lives and who calls whom.

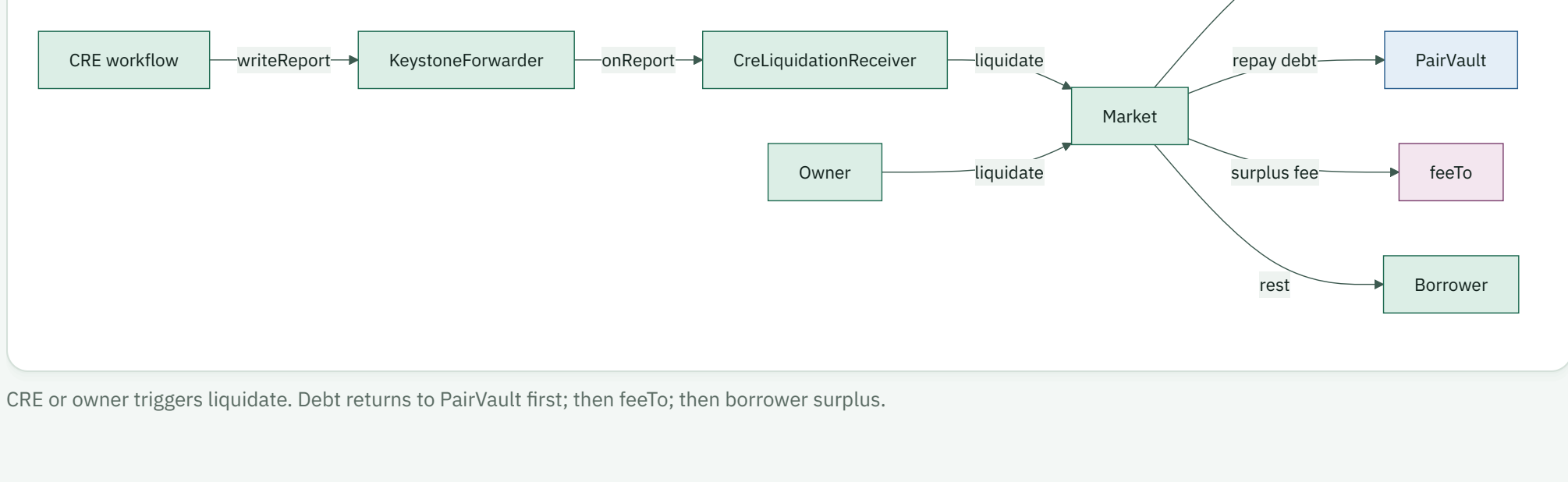
2.1 Money flow (lend / borrow)



2.2 Risk stack (what Market uses)



2.3 Liquidation (CRE + fee)

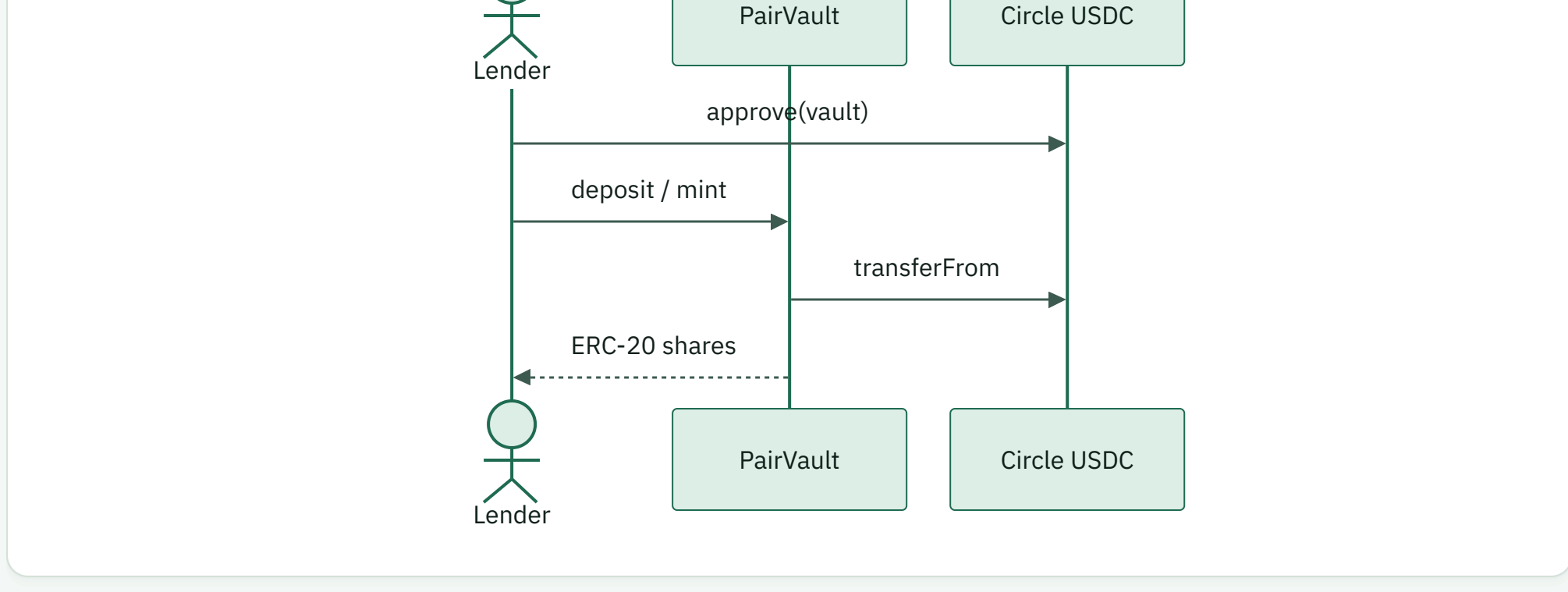


3. Who holds what

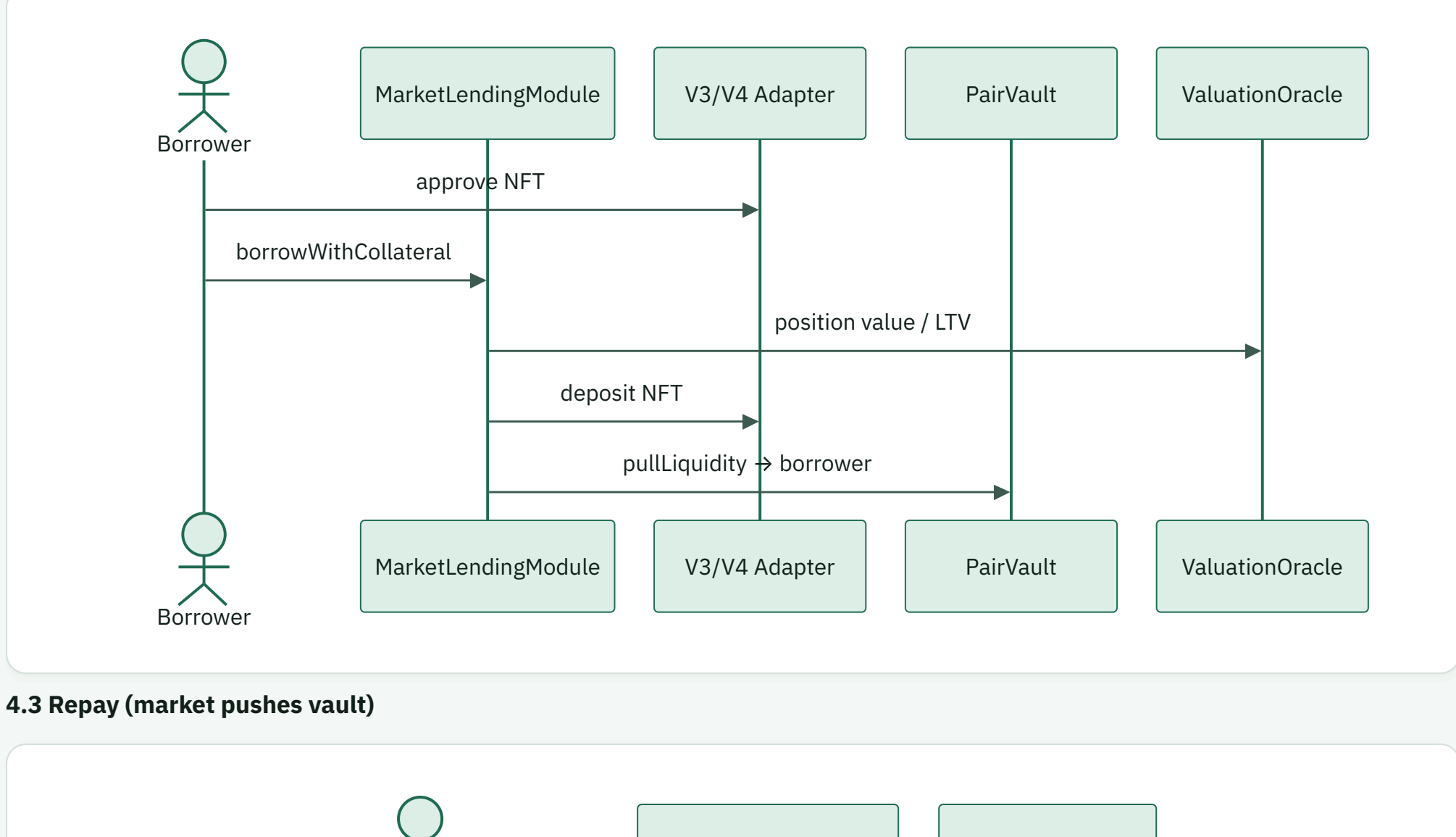
ASSET	CUSTODY	NOTES
Idle Circle USDC	PairVault (per pair)	Lender deposits / redeems here
Outstanding principal + interest	Tracked on market; owed to vault	Included in vault <code>totalAssets()</code>
DEX V3 LP NFT	V3Adapter	Unwind on liquidate
DEX V4 LP NFT	V4Adapter	Native ETH mapped to WETH for pair / valuation
Liquidation fee	Transferred to <code>feeTo</code>	Only from surplus after vault repaid

4. Core flows

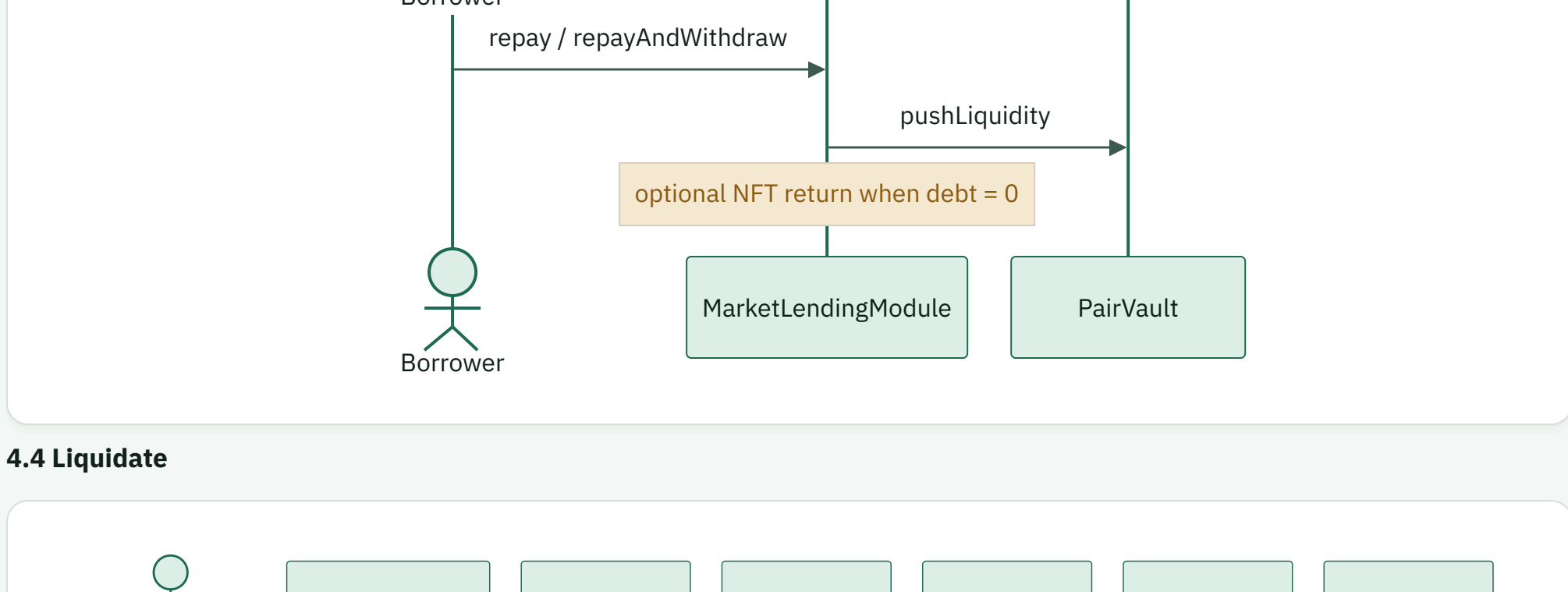
4.1 Supply (lender → vault)



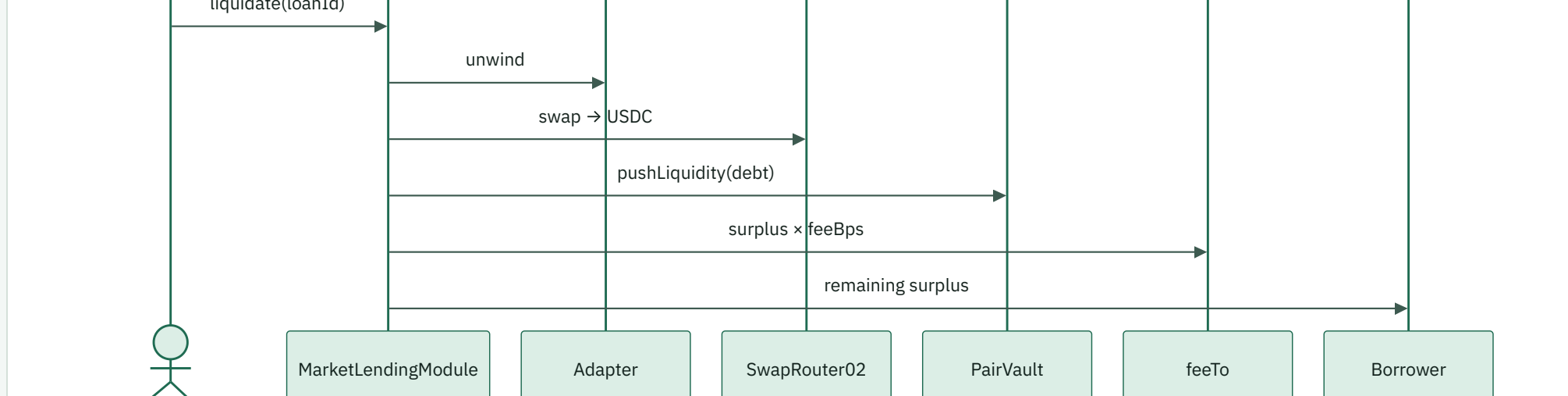
4.2 Borrow (market pulls vault)



4.3 Repay (market pushes vault)



4.4 Liquidate



5. Contracts — purpose & surfaces

ERC-4626

PairVault

smart-contracts/contracts/markets/PairVault.sol

Per-pair idle USDC vault. Standard ERC-4626 `deposit / mint / withdraw / redeem`. Only the linked market may `pullLiquidity / pushLiquidity . totalAssets = idle + market.assetsOwedToVault(pairId)`.

deposit / mintwithdraw / redeempullLiquiditypushLiquidityidleAssetstotalAssetsmaxWithdraw / maxRedeem

CORE

MarketLendingModule

smart-contracts/contracts/markets/MarketLendingModule.sol

Loans, LTV, interest accrual, liquidation, vault registry (`vaultOf`), and fee config (`feeTo` , `liquidationFeeBps`). Does not hold idle lender cash.

Borrower

`previewBorrow``borrowWithCollateral``borrowMore``repay``repayAndWithdraw``withdrawCollateral``currentDebt`

Vault accounting

`vaultOf``setVault``assetsOwedToVault``availableToBorrow`

Liquidation / health

`isLiquidatable``previewLiquidation``findFirstLiquidatableLoan``liquidate``accrueInterest`

RATES

BorrowRateConfig

contracts/markets/BorrowRateConfig.sol

Per-pair borrow APR and whether a pair is enabled.

setPairgetBorrowAprBpsisPairSupportedpairIdOf

RATES

StaticApySource

contracts/markets/StaticApySource.sol

Display lender APY only — does not move funds.

setApyBpssetDefaultApyBpsgetLenderApyBps

ORACLE

ValuationOracle

contracts/ValuationOracle.sol

LP amounts × Chainlink USD → debt-asset units for LTV.

getPositionValueUsdgetTokenPriceUsdconvertUsdToDebtAssetsetFeedsetAdapter

CUSTODY

V3Adapter

contracts/adapters/V3Adapter.sol

Holds V3 LP NFTs; `deposit / withdraw / unwind` for liquidation.

depositwithdrawunwindownsgetPositionMetagetAmountsForValuation

CUSTODY

V4Adapter

contracts/adapters/V4Adapter.sol

Same role for V4 positions; ETH → WETH mapping for pair identity.

depositwithdrawunwindregisterPositiongetAmountsForValuation

CRE

CreliquidationReceiver

contracts/markets/CreliquidationReceiver.sol

Forwarder `onReport` → `market.liquidate`. Does not configure fees.

needsUpkeeponReportsetForwarderSetMarket

AREA	ROLE
<code>IPairVault</code> / vault accounting interfaces	Pull/push + assets owed surface
<code>IPositionAdapter</code>	Adapter surface for market + oracle
<code>IValuationOracle</code>	Oracle surface
<code>TickMath</code> / <code>FullMath</code> / <code>LiquidityAmounts</code> / <code>PoolId</code>	DEX math for valuation
<code>LendingTypes</code>	Shared protocol version (V3 / V4)
<code>mocks/*</code>	Hardhat tests only

6. Fee structure (on market)

KNOB	MEANING
<code>liquidationFeeBps</code>	% of surplus after vault debt repaid (<code>1000 = 10%</code>)
<code>feeTo</code>	Fee recipient — independent of Ownable <code>owner</code>

Order on liquidate: repay PairVault → feeTo `feeTo` → remainder to borrower. Shortfall → no fee. Set via `setLiquidationFeeBps / setFeeTo` or redeploy env vars.

7. External dependencies

DEPENDENCY	USED FOR	CALLED BY
ERC20 Circle USDC	Debt asset & vault asset	PairVault, MarketLendingModule
DEX V3 NonfungiblePositionManager	Transfer / decrease / collect LP NFTs	V3Adapter
DEX V4 PositionManager + StateView	V4 custody & unwind	V4Adapter
DEX SwapRouter02	Non-USDC → USDC on liquidation	MarketLendingModule
CHAINLINK price feeds	Token USD for LTV	ValuationOracle
CRE KeystoneForwarder	Deliver signed reports	CreliquidationReceiver

8. Repo layout (contracts package)

smart-contracts/
├─ contracts/
│ └─ markets/
│ │ └─ MarketLendingModule.sol
│ │ └─ PairVault.sol
│ │ └─ CreliquidationReceiver.sol
│ │ └─ BorrowRateConfig.sol
│ │ └─ StaticApySource.sol
├─ adapters/ # V3Adapter, V4Adapter
├─ ValuationOracle.sol
├─ interfaces/
├─ libraries/
├─ mocks/
├─ scripts/ # tests only
├─ test/ # redeploy, fee setters, local keeper
└─ deployments/ # sepolia-markets.json

GIT

Repository

github.com/unblocktechie/LPFy

Key packages: `smart-contracts/` · `web/` · `cre/` · `docs/`

Deployed contracts (Sepolia)

From <code>smart-contracts/deployments/sepolia-markets.json</code> . External Uniswap / Chainlink / Circle addresses are in section 7 .	
CONTRACT	ADDRESS
MarketLendingModule	<code>0x658A244Ad0c51F0C49d1fF79678bc642365b6E65</code>
ValuationOracle	<code>0xA659A2C34c5E9826f7777AF654d0F742f8444f7F</code>
V3Adapter / V4Adapter	<code>0x8Aa1b1e3941E9b5f6b947e6213ab68f397845768</code> / <code>0x33234ac1508034A9494baa3082D0e55a6b1ae92Bd</code>
BorrowRateConfig	<code>0xb2e0c7b38b7DECE87A4E3d41A465F0175Ff65d25</code>
StaticApySource	<code>0xb20b69fEB316BCBaCc7167Dc1764637a7e8D296C2</code>
CreliquidationReceiver	<code>0xc9682A8649B7587e43a9cb85068d23081C5397F7</code>
PairVault USDC/WETH	<code>0x9478AD78af6c7758ADf8Aa78C2cc5ADEd5990843</code>
PairVault USDC/USDT	<code>0xC934C71ed024Da403f668EC4C9e26C996d7E7Ca</code>
PairVault USDC/WBTC	<code>0xab30B8fEB8Adf082094Db087Df3CaE632e27Ca0</code>